Supplementary Information

Dams mark multi-year drought vulnerability rather than buffer it in Chile's megadrought

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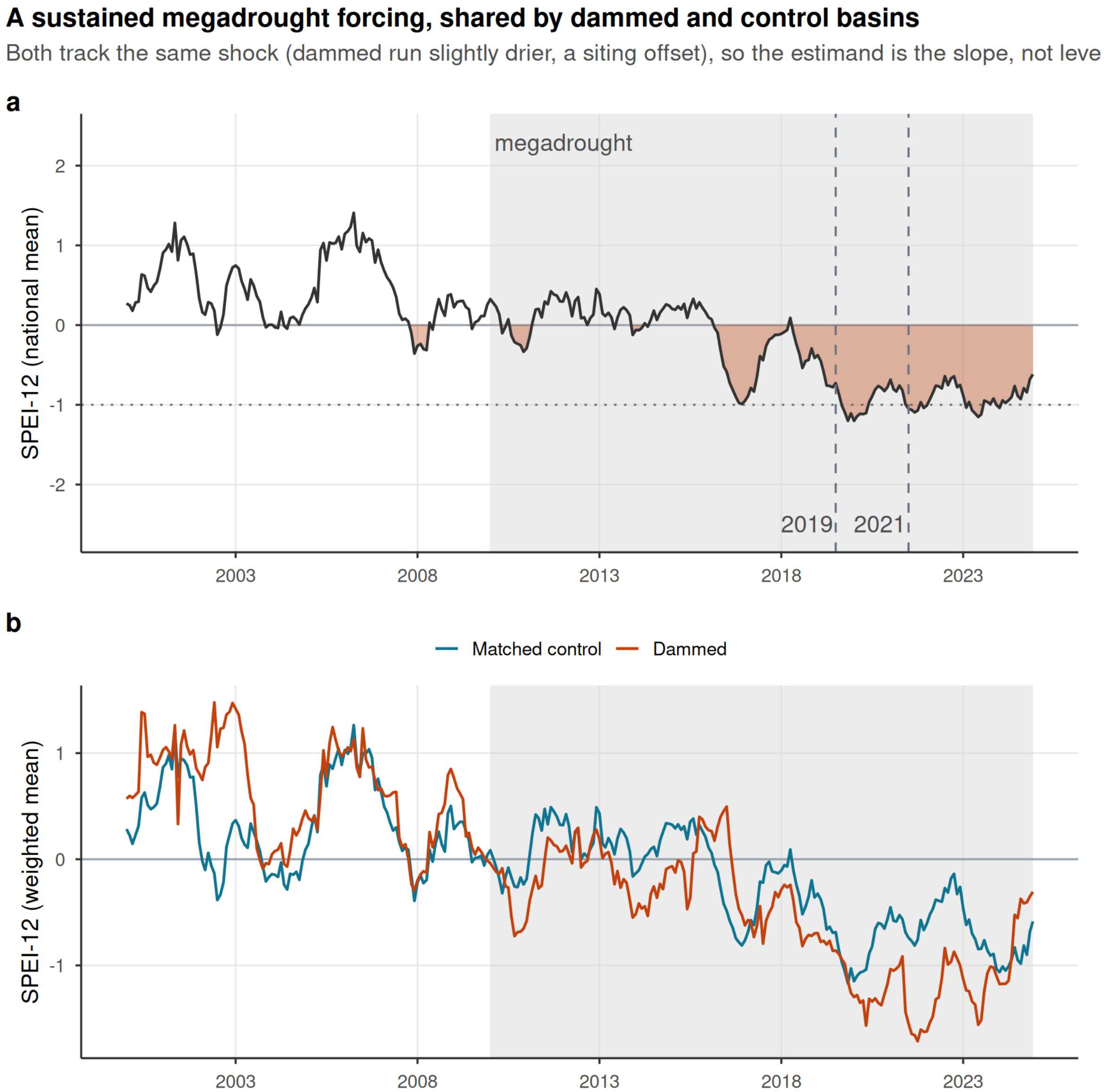
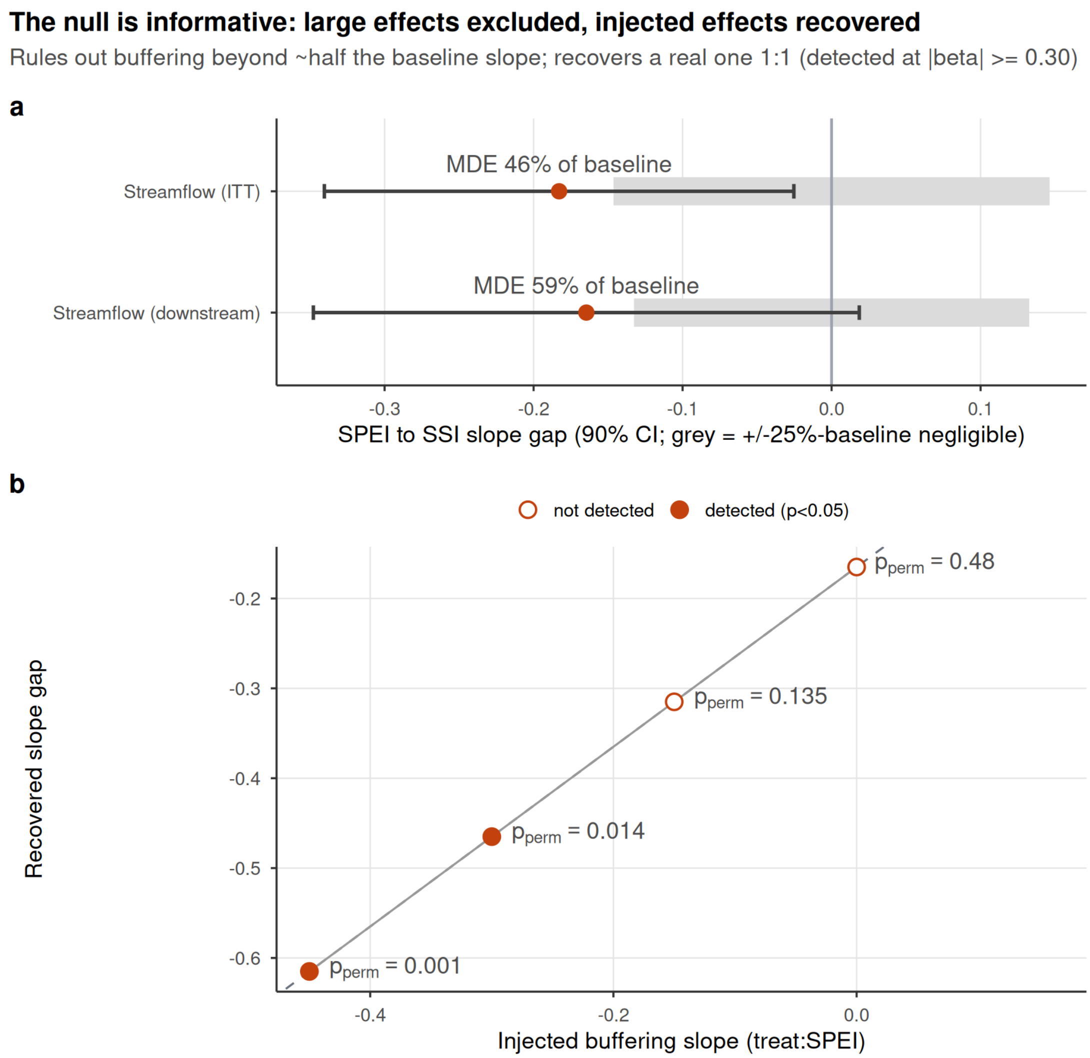
This Supplementary Information collects the supporting design, robustness, and diagnostic results referenced in the main text. Supplementary Figures S1–S3 document the drought forcing, the informativeness of the null, and the siting-confound decomposition; Supplementary Figures S4–S5 give the full study-area map and the matching diagnostics; Supplementary Figure S6 is the water-rights induced-demand test and Supplementary Figure S7 the barren-land ET-confound demonstration; Supplementary Tables S1–S6 report the equivalence bounds, placebo comparability, positive control, decomposition ladder, storage-band trends, and the aridity-overlap sensitivity; Supplementary Tables S7–S9 add the fully non-parametric overlap re-match, the winsorization-threshold sensitivity, and the spatial-autocorrelation assessment with spatially-restricted inference; Supplementary Table S10 bounds the groundwater-substitution confound with the DGA well hydrograph network; Supplementary Table S11 tests reservoir-use heterogeneity (irrigation-only subset); Supplementary Table S12 tests the snowmelt confound on the differential estimand of the upstream/downstream placebo with standardized ERA5-Land snow-water-equivalent (snow-adjusted refits, a low-snow subsample, snow-year modulation, and an early/late stability split); Supplementary Table S13 shows the matching is insensitive to recomputing baseline aridity on the strictly pre-drought 1991–2009 window; Supplementary Table S14 reports the minimum detectable effect for the cropland pre-trend; Supplementary Table S15 collects the inter-basin spillover (cuenca disjointness) and administrative-demand checks, including the treated-versus-control closure test; Supplementary Table S16 translates the equivalence margin into physical streamflow volumes; Supplementary Table S17 reports the strict hard-balanced aridity sensitivity (effective control sample 19); and Supplementary Table S18 gives the rank-based tests on raw, uncapped water-rights volumes; Supplementary Table S19 the volumetric-scale (log-flow) streamflow checks; Supplementary Table S20 the metric and functional-form checks; and Supplementary Table S21 the relegated evapotranspiration outcomes. All items are regenerated by the `targets` pipeline.

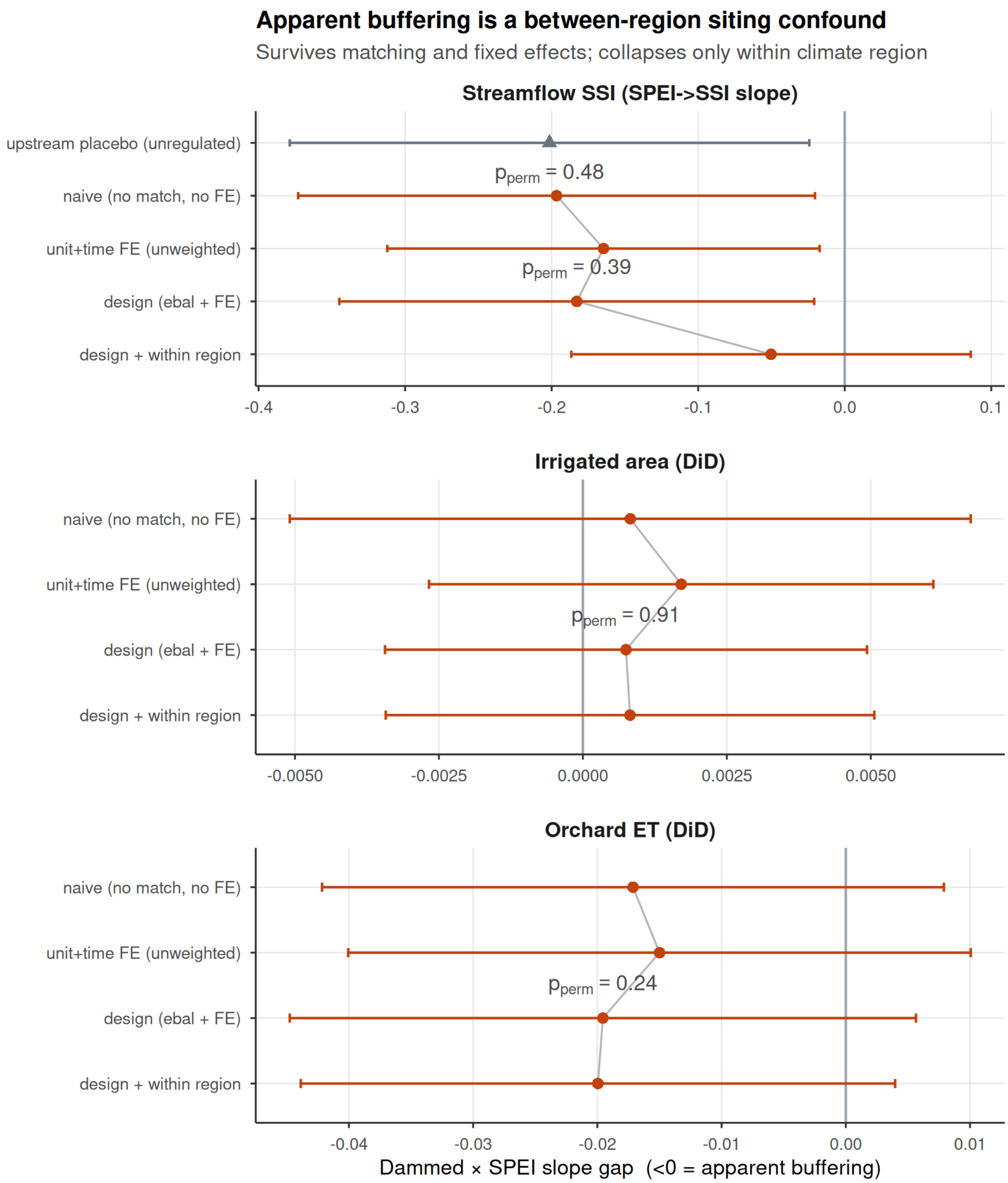
Supplementary Figures



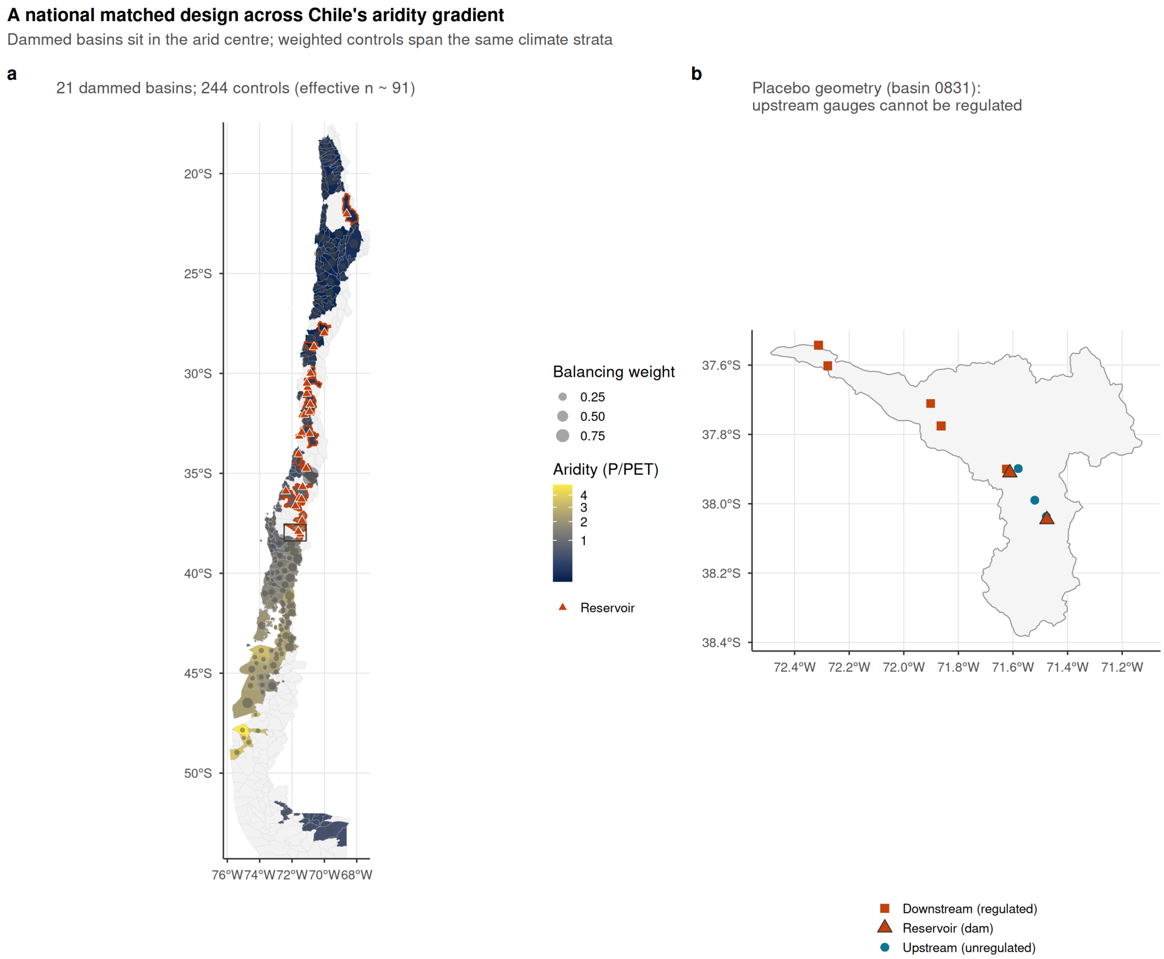
Supplementary Figure S1 | The drought forcing. (a) National mean SPEI-12 across the matched basins, 2000–2024; the negative (drought) excursion is filled and the post-2010 megadrought era shaded, with the 2019 and 2021 hyperdrought winters marked. The deficit is sustained and deep (below −1 for much of 2016–2024), giving the SPEI dose real dynamic range. (b) Entropy-balancing-weighted mean SPEI-12 for dammed versus matched-control basins: both co-move through the same megadrought (a common shock that year fixed effects absorb), with dammed basins running slightly drier, a siting offset that is precisely why the estimand is the differential deficit-to-impact slope rather than a level or calendar-time contrast.



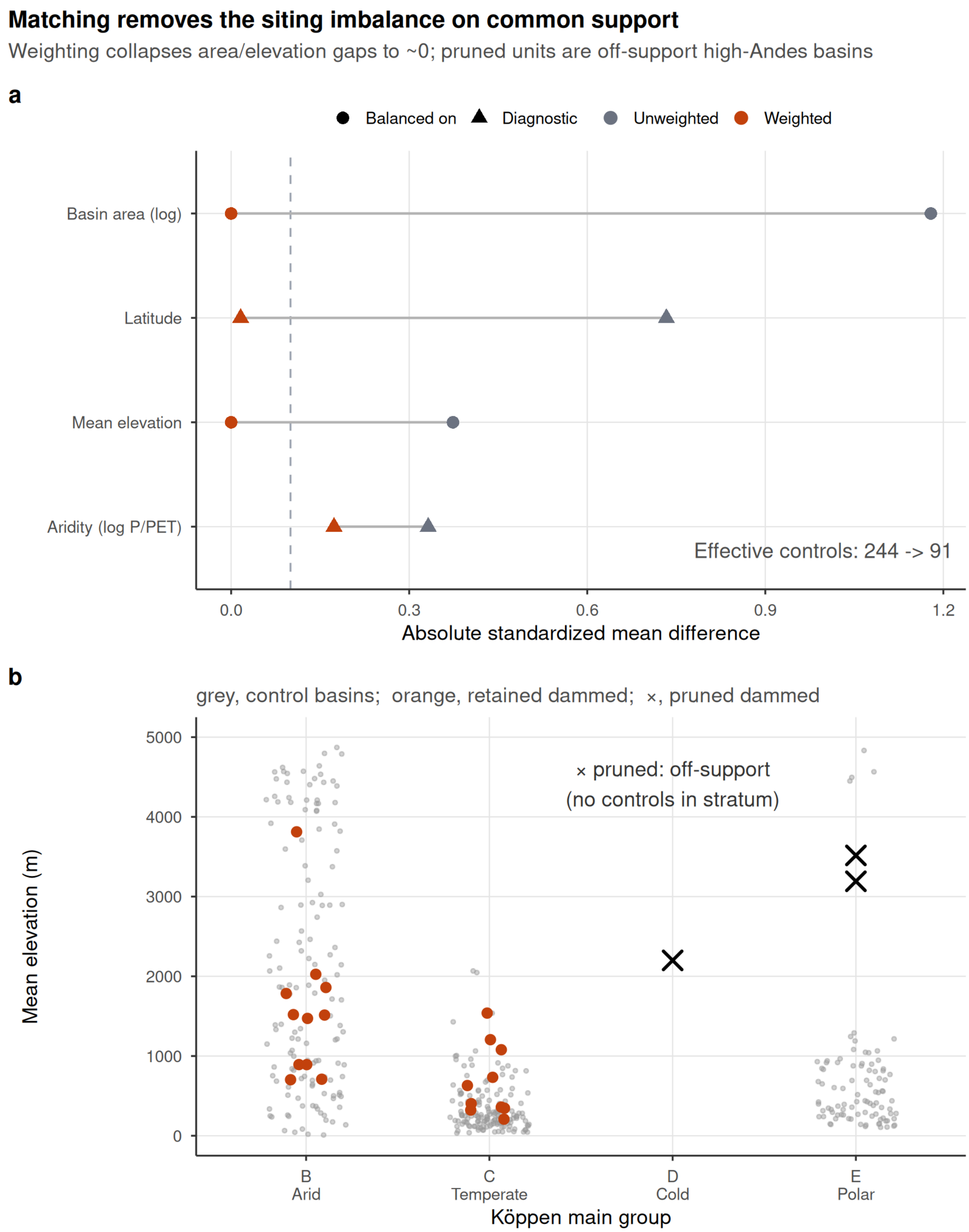
Supplementary Figure S2 | The null is informative, not underpowered blindness. (a) Equivalence / minimum-detectable-effect bounds for the two streamflow outcomes: the observed slope gap (point) and its 90% (TOST) interval against the $\pm$25%-of-baseline negligible region (grey). The 90% intervals exceed the negligible region, so formal equivalence to zero is not claimed, but they exclude attenuations larger than the minimum detectable effect (46% of baseline transmission for the intent-to-treat design, 59% downstream). (b) Positive control: a known buffering slope injected into the treated downstream gauges is recovered one-to-one by the estimator (points on the dashed expected-recovery line), the no-injection case is correctly not flagged, and randomization inference detects the effect (filled points) once the injected slope reaches -0.30. The estimator would therefore reveal a large operational buffering effect if one existed.



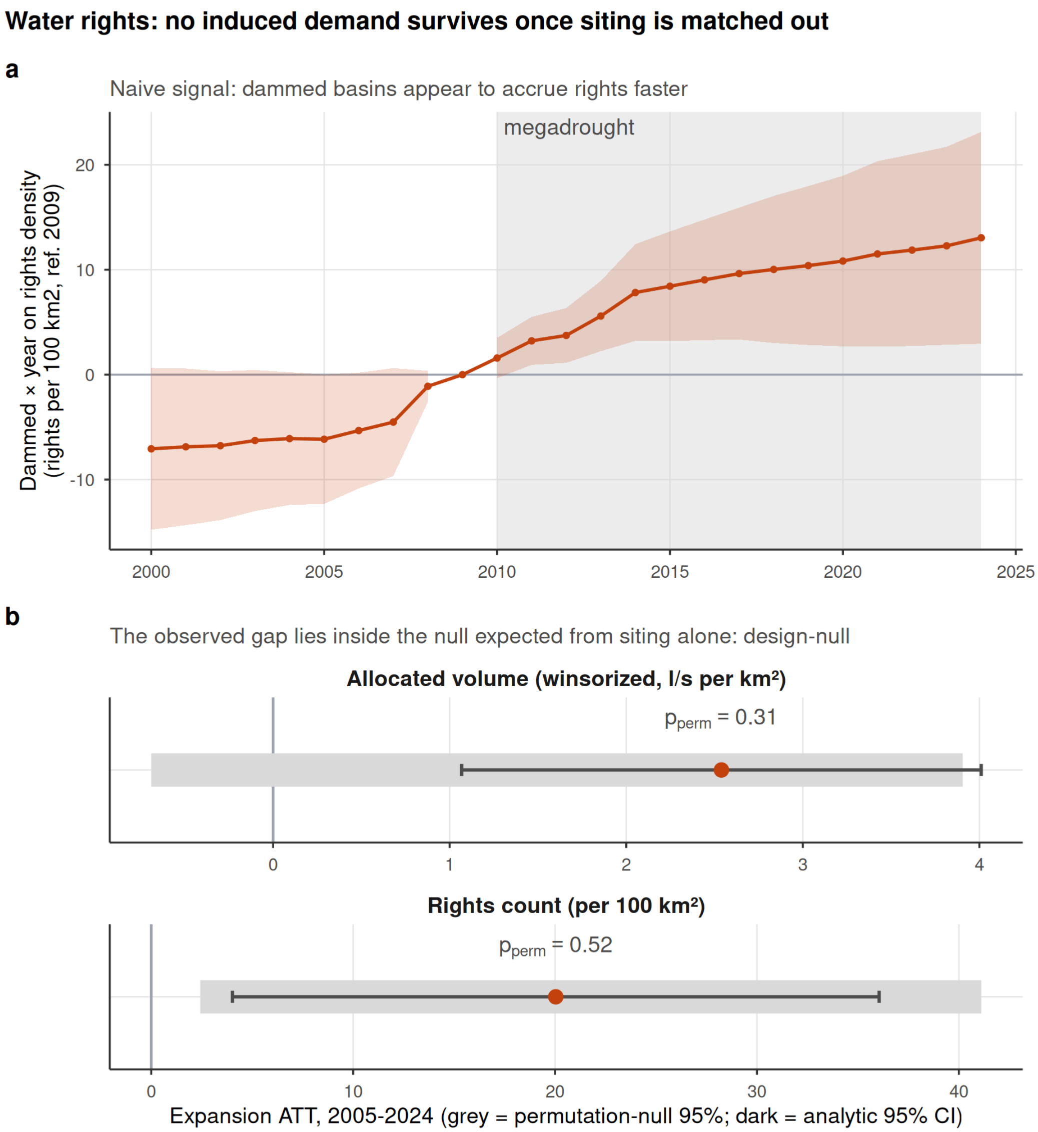
Supplementary Figure S3 | Siting-confound decomposition ladder. Dammed-versus-control transmission-slope gap (treat:SPEI; <0 = apparent buffering) estimated by increasingly stringent estimators, for the streamflow SSI slope (top) and the two difference-in-differences proxy outcomes; bars are 95% cluster-robust confidence intervals and the design and within-region rows carry the design's randomization-inference p. Rungs: (1) naive pooled (no matching, no fixed effects, what a conventional dammed-versus-undammed study reports); (2) unit and time fixed effects, unweighted; (3) the design estimator (entropy-balance weights + fixed effects); (4) design with the baseline SPEI-to-outcome slope allowed to vary within climate region. For streamflow the apparent buffering survives rungs (1) to (3) near -0.18 (permutation-null throughout), then collapses toward zero at rung (4), and is equally strong at the unregulated upstream placebo (grey triangle): the signature of a between-region aridity confound, not regulation. The proxy outcomes carry no signal at any rung.



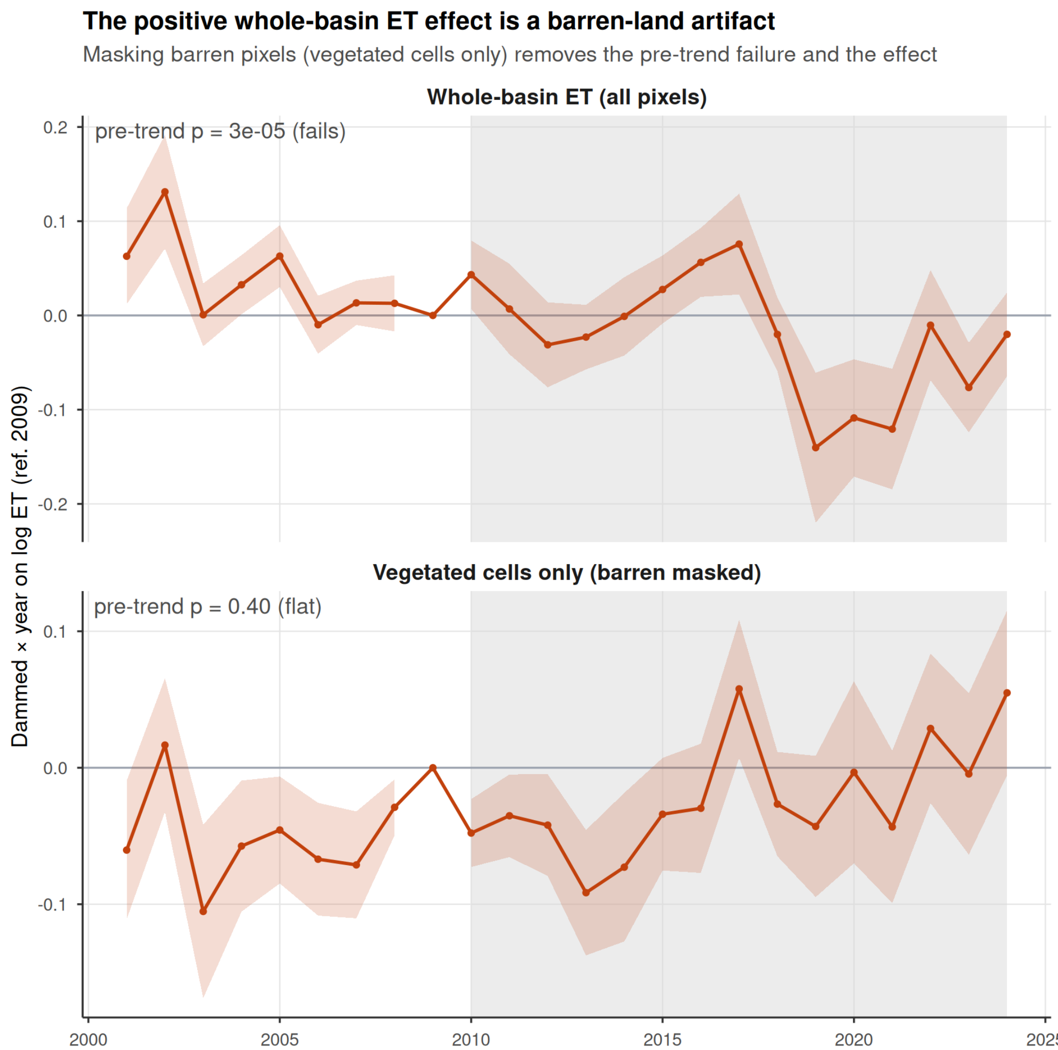
Supplementary Figure S4 | The national matched design across Chile's aridity gradient. (a) Matched sample of subcuencas, each filled by long-term aridity (P/PET, the north-south gradient the design conditions on); dammed (treated) basins are outlined in orange and marked with their reservoir (triangles), and each matched-control basin carries a centroid dot sized by its entropy-balancing weight, so the effective control sample (91) is visible against the nominal 244. Dammed basins concentrate in the arid centre and north; weighted controls span the same climate strata. (b) Within-basin placebo geometry for one representative dammed basin: the reservoir (triangle) with its downstream (regulated) gauges below the dam and upstream (unregulated) gauges above it.



Supplementary Figure S5 | Covariate balance and common support for the matched design. (a) Love plot of the absolute standardized mean difference between dammed and control basins before (unweighted) and after entropy balancing, for the two balanced covariates (log basin area, mean elevation) and two untargeted diagnostics (latitude, log aridity); the dashed line is the conventional 0.1 balance threshold. Weighting collapses the large siting imbalances to ~0, leaving a small residual aridity difference (0.17) that the doubly-robust outcome model absorbs, and reduces the effective control sample from 244 to 91. (b) Climate-stratum common support: mean elevation of control (grey) and dammed (orange, retained) basins within each Köppen main group; the three pruned dammed basins (x) fall in the cold (D) and polar (E) strata, whose controls lie far below them, so they are off-support and excluded rather than matched by extrapolation.



Supplementary Figure S6 | Water rights show no induced demand once siting is matched out. (a) Event study of cumulative consumptive water-rights density (dammed x year, ref. 2009; rights per 100 km2): the dammed-versus-control gap appears to widen through the megadrought, the naive induced-demand signal. (b) For both the rights count and the winsorized allocated volume, the expansion ATT (2005-2024; orange point, dark analytic 95% CI) is positive and its analytic interval excludes zero, yet it lies squarely inside the 95% band of the permutation null (grey), what random treatment assignment within climate x aridity strata already produces given the dammed basins' markedly higher aridity, so it is null under randomization inference (p_perm = 0.52 for counts, 0.31 for volume). A conventional matched analysis of this registry would report induced demand; the design-based inference reveals it as siting.



Supplementary Figure S7 | The positive whole-basin ET effect is a barren-land artifact. Event studies of the dammed-versus-control evapotranspiration response (dammed x year, ref. 2009) for (a) whole-basin ET over all pixels, which shows strongly non-parallel pre-trends (pre-trend slope p = 3x10^-5) and an apparent post-2010 divergence, and (b) ET restricted to vegetated irrigated (orchard) cells, i.e. with barren and salar pixels masked out. Masking the barren land removes both the pre-trend failure (p = 0.40) and the positive effect (the forcing-conditioned slope gap moves from +0.048 to -0.020), confirming that the exclusion of whole-basin ET from the main results is mechanistic (MOD16 ET is contaminated over sparsely vegetated barren surfaces), not a post-hoc drop of an inconvenient result.

Supplementary Tables

Supplementary Table S1 | Equivalence / minimum-detectable-effect bounds on the headline nulls. Observed slope gap, MDE (smallest effect detectable at 80% power from the permutation-null SD), the 90% (TOST) interval, and whether it lies within the negligible region (±25% of baseline). "baseline" is the untreated (control) SPEI→SSI / SPEI→impact transmission slope from the fixed-effects model. No outcome is formally equivalent to zero; the streamflow design rules out attenuations larger than ~46–58% of baseline transmission.

Outcome	Observed	MDE	90% CI	Equiv. to 0?	p_perm
streamflow SSI (ITT)	-0.183	0.268	[-0.34, -0.0253]	no	0.39
streamflow SSI (downstream)	-0.165	0.312	[-0.348, 0.0186]	no	0.48
irrigated area (DiD)	0.001	0.010	[-0.00506, 0.00656]	no	0.90
orchard ET (DiD)	-0.020	0.046	[-0.0463, 0.00718]	no	0.25

Supplementary Table S2 | Upstream/downstream placebo comparability. Upstream gauges are higher than downstream, and gauge elevation predicts the transmission slope among undammed controls (−0.21 per +1000 m), yet upstream and downstream treated gauges have nearly identical raw transmission slopes, so the equal up/down attenuation is not an elevation artifact.

quantity	value	detail
upstream gauges (elev, mean m)	1003.000	n=28; mean transmission 0.565
downstream gauges (elev, mean m)	459.000	n=40; mean transmission 0.599
control transmission slope vs elevation (per +1000 m)	-0.212	p = 0.00 (n=84 control gauges); elevation predicts transmission, so the up/down gap is bounded with this sensitivity rather than assumed away

Supplementary Table S3 | Positive control on the streamflow estimator. A known buffering slope (`beta_injected`; negative = buffering) is added to treated downstream units; the estimator recovers it (recovered = baseline −0.17 + beta), the no-effect case is not flagged, and randomization inference detects the effect once it exceeds the minimum detectable effect.

beta_injected	recovered	p_perm	detected
0.00	-0.165	0.480	FALSE
-0.15	-0.315	0.135	FALSE
-0.30	-0.465	0.014	TRUE
-0.45	-0.615	0.001	TRUE

Supplementary Table S4 | Siting-confound decomposition ladder (numeric). The dammed-versus-control transmission-slope gap at each rung of Supplementary Figure S3, for streamflow SSI (ITT) and the two difference-in-differences proxy outcomes, plus the within-basin upstream placebo. `perm_p` is the design's randomization-inference p for the design rung and the placebo.

Outcome	Rung	Slope gap	SE	p_perm
Streamflow SSI (SPEI->SSI slope)	naive (no match, no FE)	-0.197	0.090	-
Streamflow SSI (SPEI->SSI slope)	unit+time FE (unweighted)	-0.165	0.075	-
Streamflow SSI (SPEI->SSI slope)	design (ebal + FE)	-0.183	0.083	0.39
Streamflow SSI (SPEI->SSI slope)	design + within region	-0.050	0.070	-
Irrigated area (DiD)	naive (no match, no FE)	0.001	0.003	-
Irrigated area (DiD)	unit+time FE (unweighted)	0.002	0.002	-
Irrigated area (DiD)	design (ebal + FE)	0.001	0.002	0.91
Irrigated area (DiD)	design + within region	0.001	0.002	-
Orchard ET (DiD)	naive (no match, no FE)	-0.017	0.013	-
Orchard ET (DiD)	unit+time FE (unweighted)	-0.015	0.013	-
Orchard ET (DiD)	design (ebal + FE)	-0.020	0.013	0.24
Orchard ET (DiD)	design + within region	-0.020	0.012	-
Streamflow SSI (SPEI->SSI slope)	upstream placebo (unregulated)	-0.201	0.090	0.48

Supplementary Table S5 | Storage-band trends and megadrought level shift. Top: per-year trend slope of the cross-reservoir peak, trough, and peak-trough amplitude (percent of capacity per year; reservoir fixed effects, cluster-robust). Peak and trough both decline while the amplitude trend is not distinguishable from zero, consistent with a supply-side level decline rather than a loss of refill capacity. Bottom: because the storage dynamics are non-linear over a window that begins just before the megadrought, the linear slope is reported as a summary rate only, and the decline is additionally characterized as a within-reservoir level shift at the 2010 onset (pre versus post means with reservoir fixed effects): peak and trough step down significantly while the amplitude shift is indistinguishable from zero.

Component	Trend (%/yr)	95% CI	p
peak	-1.281	[-2.21, -0.349]	0.013
trough	-1.220	[-1.72, -0.72]	0.000
amplitude	-0.061	[-1.13, 1.01]	0.912

Component	Pre-2010 mean	Post-2010 mean	Shift	95% CI	p
peak	0.867	0.665	-0.210	[-0.364, -0.0555]	0.01400
trough	0.395	0.248	-0.150	[-0.227, -0.0734]	0.00081

Component	Pre-2010 mean	Post-2010 mean	Shift	95% CI	p
amplitude	0.473	0.417	-0.059	[-0.217, 0.0987]	0.47000

Supplementary Table S6 | Aridity-overlap sensitivity. Key results re-estimated on the sub-sample of basins whose baseline log-aridity lies in the treated/control overlap band (all 21 treated basins and 158 of 244 controls fall inside it), to show the parametric aridity adjustment is not masking a true effect where common support is genuine. The streamflow transmission-slope gap is essentially unchanged, and the water-rights expansion ATT shrinks and now straddles zero.

Quantity	Full sample	Aridity overlap	Overlap 95% CI
Streamflow SPEI->SSI slope gap	-0.183	-0.174	-
Water-rights expansion ATT	20.036	11.880	[-5.62, 29.4]

Supplementary Table S7 | Fully non-parametric matching on the aridity-overlap subset. Every cross-sectional outcome re-estimated after re-fitting entropy balancing *within* the treated/control aridity-overlap band and using the weighting-only estimator (ebal-weighted difference in means, no outcome-model aridity term), so the estimate performs no parametric extrapolation across the aridity regime gap. Cropland-area expansion, orchard-ET buffering and the water-rights count remain null. The winsorized water-rights volume is the only outcome whose stratified-permutation p falls below 0.05 here; Supplementary Table S9 shows this reflects spatial autocorrelation, not a reservoir effect. The `perm p` column is the unit-level stratified randomization-inference p (Köppen × aridity tercile).

Outcome	ATT	95% CI (HC3)	p_perm	n_t	n_c
Cropland-area expansion (ha/km2)	-0.5860	[-2.08, 0.905]	0.328	21	158
Orchard ET buffering (dlogET/dSPEI)	-0.0182	[-0.0427, 0.00631]	0.291	18	30
Water-rights count (per 100 km2)	12.0000	[-4.36, 28.4]	0.420	21	136
Water-rights volume (l/s per km2)	1.8700	[0.164, 3.58]	0.005	21	136

Supplementary Table S8 | Sensitivity of the water-rights volume null to the winsorization threshold. The winsorized allocated-volume expansion ATT and its randomization-inference p, with individual consumptive flows capped at the 95th, 99th, 99.5th (the value used in the main text) and 99.9th percentiles of the consumptive-flow distribution. The analytic interval excludes zero at every threshold, but the effect is not significant under randomization inference at any threshold (permutation p from 0.39 to 0.21), so the volume null is not an artefact of the winsorization choice. The cap is also physically validated (second table): the 99.5th-percentile cap corresponds to 0.3 m^3 s^{-1}, and in every gauged basin the largest retained right is below the basin's maximum observed monthly mean streamflow (median ratio 0.002), so no physically impossible values survive the cap; the uncapped registry maximum, by contrast, exceeds 2×10^8 l/s.

Winsor pctile	Volume ATT	95% CI	p_perm
95.0	1.837	[0.639, 3.03]	0.389
99.0	2.395	[0.956, 3.83]	0.343
99.5	2.539	[1.07, 4.01]	0.310
99.9	3.211	[1.56, 4.86]	0.209

Quantity	Value	Detail
winsorization cap (99.5th pct of consumptive flows)	287.000	l/s (= 0.3 m3/s); uncapped registry maximum 2.08e+08 l/s
gauged basins where the largest retained right <= max observed monthly flow	77.000	of 77 gauged basins (100%)
median ratio: largest retained right / max observed flow	0.002	IQR [0.000, 0.004]

Supplementary Table S9 | Spatial autocorrelation and spatially-restricted inference. Top: Moran's I of each outcome's ATT-model residuals (great-circle distances, five-nearest-neighbour row-standardized weights, 999 residual permutations). The water-rights outcomes and cropland area show significant positive spatial autocorrelation, which the unit-level stratified permutation does not preserve. Bottom: for the water-rights outcomes, the ATT with its spatially-naive (HC3) interval and stratified-permutation p alongside the cuenca-block-clustered interval and the spatial-block permutation p (treatment swapped at the cuenca level). The apparent water-rights volume effect on the overlap subset (stratified p = 0.005) is not significant once spatial dependence is respected (spatial-block p = 0.077; cuenca-clustered interval spans zero), and the count is null under both.

Outcome	n	Moran's I	Expected	p
Cropland-area expansion	265	0.181	-0.004	0.001
Water-rights count	216	0.599	-0.005	0.001
Water-rights volume	216	0.388	-0.005	0.001
Orchard ET buffering	49	0.069	-0.021	0.198

Outcome	ATT	95% CI (HC3)	p (stratified)	95% CI (cuenca)	p (spatial block)
Water-rights volume (l/s per km2)	1.87	[0.16, 3.58]	0.005	[-0.33, 4.07]	0.077
Water-rights count (per 100 km2)	12.00	[-4.36, 28.36]	0.420	[-2.70, 26.70]	0.581

Supplementary Table S10 | Groundwater-substitution bound from the DGA well hydrograph network. Differential groundwater drawdown between dammed and matched-control basins over the megadrought (2010–2021). Each well's depletion rate is the OLS slope of Hampel-filtered depth to water on time (m yr^{-1}, positive = falling water table); per-basin rates are aggregated by the outlier-robust median (primary) and, for robustness, the mean. `dammed` / `control` are the weighted group-mean rates; `att_wo` is the entropy-balancing weighting-only ATT (with 95% CI and stratified randomization-inference `perm_p`) and `att_dr` the aridity-adjusted doubly-robust ATT. Coverage is 26 matched basins (13 dammed, 13 control, 213 wells). The differential is not significant in either specification, and the depletion residuals carry no spatial autocorrelation (`moran_I`, `moran_p`), so dammed basins show no faster groundwater drawdown of the kind a substitution confound would require. One scope limitation applies: the DGA monitoring network is not designed around agriculture, so monitored wells need not sit inside the high-value orchard footprints where compensatory pumping concentrates, nor sample the same aquifer strata. The basin-median rates therefore bound basin-scale substitution; field-scale pumping localized outside the monitored network remains a residual caveat.

Spec	Dammed (m/yr)	Control (m/yr)	ATT (WO)	95% CI	p_perm	ATT (DR)	Moran I	Moran p	wells
median (primary)	0.098	0.079	0.089	[-0.17, 0.35]	0.681	0.028	-0.069	0.752	213
mean (robustness)	0.289	0.076	0.283	[-0.11, 0.67]	0.328	0.486	-0.014	0.726	213

Supplementary Table S11 | Reservoir-use heterogeneity: irrigation-only robustness. The treated sample is dominated by irrigation reservoirs (18 of the 21 matched treated basins hold one; three are purely hydropower or potable supply). The headline streamflow SPEI-to-SSI slope gap (`treat:SPEI`, < 0 = apparent buffering) is re-estimated on all treated basins and on the irrigation-only subset (non-irrigation treated basins dropped, controls unchanged), with the design's stratified randomization-inference p. The slope gap and its permutation-null status are unchanged, so the null is not an artefact of pooling hydropower with irrigation reservoirs; `n_treat` columns give the treated basins with streamflow gauges in each case.

Quantity	All treated	p (all)	Irrigation only	p (irrig)	n_treat (all)	n_treat (irrig)
Streamflow slope gap (ITT)	-0.183	0.388	-0.192	0.432	17	16
Streamflow slope gap (upstream placebo)	-0.201	0.482	-0.221	0.456	15	14

Supplementary Table S12 | Snowmelt-confound tests on the upstream/downstream placebo, run on the differential estimand. Upstream reaches are snow-dominated Andean headwaters, so snowmelt storage and release could in principle shape the treated-versus-control transmission differential up- versus downstream independently of regulation. All snow terms use ERA5-Land snow-water-equivalent (SWE) standardized across units, since the product underestimates absolute Andean snowpack. Four tests bound the concern: a forcing-by-snowpack adjustment leaves the up/down equivalence intact while resolving a detectable cross-sectional snow term upstream, so the product is not blind to snow; the equivalence persists among low-snow units (below-median peak SWE); the apparent buffering does not strengthen in high-snow years (snow-year modulation); and the treated-versus-control differential is statistically indistinguishable between the early and late halves of the panel (period interaction p = 0.46 upstream, 0.52 downstream), providing no evidence of a snow buffer degrading in parallel with the megadrought. Gauge-level rows cap any snow-induced up-minus-down slope difference at 0.007 against the observed 0.04; these rows are restricted to placebo-basin gauges with SWE coverage, so gauge counts differ slightly from Supplementary Table S2. Snowmelt is therefore not the driver of the placebo result.

Quantity	Value	Detail
upstream placebo: treat x SPEI, unadjusted	-0.2014	SE 0.0904, p = 0.030; 15 treated / 44 control units
upstream placebo: treat x SPEI, snow-adjusted (spei x peak-SWE z)	-0.1659	SE 0.0899, p = 0.070; snow term 0.0922 (SE 0.0360, p = 0.013)
upstream placebo: treat x SPEI, low-snow units (below-median peak SWE)	-0.0829	SE 0.0791, p = 0.304; 3 treated / 26 control units; peak SWE 0.0000-0.0012 m w.e.
upstream placebo: treat x SPEI, early vs late period	-0.3207	early (<=2010) -0.3207 (SE 0.1180, p = 0.009); late (>2010) -0.2654 (SE 0.1795, p = 0.145); period change (treat x SPEI x late) 0.1034 (SE 0.1394, p = 0.461)
upstream placebo: treat x SPEI x annual-SWE anomaly (snow-year modulation)	-0.0313	SE 0.0472, p = 0.510; base spei x snow-year 0.0521 (SE 0.0374, p = 0.169)
downstream: treat x SPEI, unadjusted	-0.1646	SE 0.0851, p = 0.058; 15 treated / 44 control units
downstream: treat x SPEI, snow-adjusted (spei x peak-SWE z)	-0.1413	SE 0.0921, p = 0.131; snow term 0.0296 (SE 0.0262, p = 0.264)
downstream: treat x SPEI, low-snow units (below-median peak SWE)	-0.0808	SE 0.1207, p = 0.509; 6 treated / 23 control units; peak SWE 0.0000-0.0005 m w.e.
downstream: treat x SPEI, early vs late period	-0.2060	early (<=2010) -0.2060 (SE 0.1276, p = 0.112); late (>2010) -0.3216 (SE 0.1851, p = 0.088); period change (treat x SPEI x late) -0.1047 (SE 0.1611, p = 0.518)
downstream: treat x SPEI x annual-SWE anomaly (snow-year modulation)	-0.1031	SE 0.0809, p = 0.207; base spei x snow-year 0.0345 (SE 0.0328, p = 0.297)

Quantity	Value	Detail
gauges: up vs down peak SWE (m w.e., ERA5-Land)	0.0682	down 0.0482 (gap 0.0200 m = 0.13 SD); n_up=28, n_down=32
control gauges: transmission slope per SD peak SWE	0.0000	SE 0.0282, 95% CI [-0.0553, 0.0553], p = 0.999 (n=82); net of elevation p = 0.808
conservative bound: snow-induced up-down slope difference	0.0072	SWE gap 0.13 SD x CI-upper |sensitivity| 0.0553; observed differential gap -0.04

Supplementary Table S13 | Pre-drought aridity-window sensitivity of the matching. The 1991–2020 WMO normal used for baseline aridity overlaps eleven megadrought years, so the covariate could in principle absorb post-treatment variation. Recomputing unit aridity on the strictly pre-drought 1991–2009 window leaves the design unchanged: unit aridity is essentially identical across windows (Pearson and Spearman rows), 99% of matched units keep the aridity tercile used as a permutation stratum, and the megadrought change in P/PET (2010–2020 minus 1991–2009) does not differ significantly between treated and weighted control basins and is an order of magnitude smaller than the treated-control aridity gap the matching addresses. The matching therefore does not lean on drought-period or reservoir-influenced variation.

Quantity	Value	Detail
unit aridity, pre-drought (1991-2009) vs full (1991-2020) window: Pearson r	0.9982	Spearman rho 0.9993 (n=265 matched units)
aridity-tercile agreement between windows (permutation strata)	0.9920	share of matched units assigned the same tercile
median P/PET, treated: pre-drought vs full window	0.2730	full 0.227; control pre 0.798, full 0.680
megadrought change in P/PET (2010-2020 minus 1991-2009), treated minus control	-0.0441	SE 0.0265, p = 0.10 (ebal-weighted); small against the ~0.5 treated-control aridity gap

Supplementary Table S14 | Minimum detectable effect for the cropland pre-trend. The flat pre-2010 differential cropland trend (Fig. 4b) is held to the same equivalence standard as the headline nulls: the minimum detectable pre-window trend slope is built from the permutation-null SD (80% power, two-sided alpha 0.05) and cumulated over the 2000–2009 pre-window as a share of the weighted control cropland level. The design would detect a pre-existing divergence accumulating to about half of the control cropland level by 2009; the observed differential trend is about a seventh of the minimum detectable slope.

Quantity	Value	Detail
pre-2010 differential cropland trend (treat:year, basin fraction per yr)	-0.000195	perm p = 0.72 (1000 perms); window 2000-2009
pre-trend MDE (80% power, alpha = 0.05, basin fraction per yr)	0.001400	permutation-null SD 0.0005
cumulative detectable pre-2010 divergence (% of control cropland level)	51.300000	MDE x 9 yr against control weighted mean fraction 0.0245

Supplementary Table S15 | Inter-basin spillover and administrative-demand checks. Top rows: matched treated and effective-control basins occupy entirely disjoint DGA cuencas, so no matched pair shares a river network and the within-cuenca contamination rule already removes the direct transfer pathway. The residual transfer bias is bounded in both directions: a transfer moving water from a dammed basin to a control raises the control's availability and shrinks the estimated gap, biasing toward the null we report, while imports sustaining a treated basin during peak drought would flatten the treated transmission slope where deliveries land, downstream, and so mimic regulation-specific buffering; the absence of any downstream-specific flattening in the placebo bounds this direction too, though a vulnerability signal partially offset by imports cannot be fully excluded without an explicit transfer-network model. Bottom rows: the water-rights registry was not closed during the megadrought, and the closure that would matter, a freeze specific to dammed basins, is absent: treated basins entered the drought with a higher allocated stock per unit area than their weighted controls, accrued new rights about 2.7 times faster through the megadrought, and none went without new grants, while 29% of weighted controls did. The demand-side null is therefore a genuine null difference, not the forced silence of saturated registries.

Metric	Value	Detail
treated cuencas	14.00	distinct DGA cuencas holding >=1 treated subcuenca
effective-control cuencas	70.00	distinct DGA cuencas holding >=1 effective (w>0) control
cuencas shared by a matched pair	0.00	contamination rule removes same-cuenca controls => no matched pair shares a river network
new consumptive rights per megadrought year	4687.00	range 2539-7352 over 2010-2021; registry not closed
basins granted new rights per year	194.00	of 265 matched basins
pre-drought (2009) rights stock, treated (per 100 km2, weighted)	15.50	control 6.2; higher stock over lower renewable supply => treated utilization at least as high
pre-drought (2009) winsorized volume stock, treated (l/s per km2, weighted)	1.84	control 0.65
megadrought accrual rate, treated (rights per 100 km2 per yr, weighted)	1.72	control 0.63; treated accrual not slower => no differential administrative closure
treated basins with zero new rights over the megadrought (share)	0.00	control share 0.29

Supplementary Table S16 | Physical-volume translation of the equivalence margin. The ±25% equivalence margin corresponds at the worst observed deficit to a perturbation of 0.29 SSI units (Methods). Translated through each treated downstream gauge's within-gauge linear sensitivity of the 12-month rolling mean flow to its SSI, that perturbation corresponds to the volumes below, expressed per 12-month season and as a share of mean annual flow. The linear sensitivity is an average rather than a dry-tail mapping and therefore overstates the worst-drought volume; even so, the median is of order tens of cubic hectometres, a volume that is material for local water management. This is why the margin is anchored to the operational drought-severity classification rather than to volumetric negligibility, why the threshold-free minimum detectable effect is reported alongside, and why the causal weight of the null rests on the within-basin placebo rather than on the equivalence verdict.

Quantity	Value	Detail
equivalence-margin volume, treated downstream gauges (hm3 per 12-month season)	58.0	median across 40 gauges; IQR [16.3, 137.2]; flow-SSI sensitivity x 0.29 SSI x 31.5; linear (average) sensitivity overstates the dry-tail volume
equivalence-margin volume as % of mean annual flow	10.7	IQR [6.9, 14.0]%

Supplementary Table S17 | Hard-balanced aridity sensitivity. The primary design deliberately leaves a residual aridity imbalance (SMD 0.17) to the doubly-robust outcome model because hard-balancing aridity collapses the effective control sample from 91 to about 19. Here the strict alternative is reported: entropy balancing that targets log baseline aridity alongside log area and elevation within Köppen group, achieving SMD 0.000 with no parametric extrapolation across the hydroclimatic regime gap. Every headline result is unchanged or strengthened: the streamflow slope gap and the upstream placebo collapse essentially to zero, and the cropland and water-rights outcomes remain null. That even the *apparent* buffering vanishes when aridity is fully balanced is the direct signature of a siting confound rather than a reservoir effect.

Quantity	Value	Detail
effective control sample size (aridity hard-balanced)	18.900000	log-aridity SMD after balancing 0.000 (residual 0.17 in the primary design)
streamflow treat x SPEI, ITT	-0.023000	SE 0.160, perm p = 0.930 (1000 perms); primary design -0.183
streamflow treat x SPEI, upstream placebo	-0.047000	SE 0.154, perm p = 0.882; primary design -0.20
cropland-area DiD slope gap	0.000508	SE 0.0023, perm p = 0.855; primary design perm p = 0.91
water-rights expansion ATT (weighting-only, per 100 km2)	-4.900000	95% CI [-36.9, 27.1], perm p = 0.855

Supplementary Table S18 | Rank-based tests on raw, uncapped water-rights volumes. Rights volumes are extremely concentrated and carry known data errors, which is why the level analysis winsorizes; rank-based statistics are immune to caps, outliers, and any monotone error in extreme caudal entries. The per-right volume distribution is nearly identical in dammed and control basins at the median, so the count density does not systematically understate demand in dammed basins. The naive basin-level rank gap in raw added volume reproduces the siting gap, as with the count; the design contrast (ebal-weighted mean-rank difference) appears significant under unit-level permutation, which is anti-conservative given the strong spatial autocorrelation of water-rights outcomes (Supplementary Table S9), and is not significant under the governing cuenca-block permutation, matching the winsorized-volume verdict.

Quantity	Value	Detail
volume per right, raw (median l/s): dammed vs control	2.000	control 1.8; Wilcoxon rank-sum p = 0.00 (n = 77,287 rights)
raw volume added 2005-2024, uncapped (median l/s per km2): dammed vs control	2.310	control 0.17; naive Wilcoxon p = 0.00 (265 basins) reproduces the siting gap the design removes
design rank contrast on raw volume (mean-rank difference, ebal-weighted)	84.700	unit-level stratified permutation p = 0.001, anti-conservative under the spatial autocorrelation of Supp Table S9
design rank contrast under spatially-restricted (cuenca-block) permutation	84.700	cuenca-block permutation p = 0.145 (1999 perms), the governing inference for water-rights outcomes; rank scale immune to caudal errors and caps
share of total raw volume held by the largest 1% of rights	0.997	raw registry (data errors included); why the level analysis winsorizes

Supplementary Table S19 | Volumetric-scale streamflow checks. SSI standardizes each gauge on its own, for regulated gauges post-dam, flow distribution, so the transmission slope tests distribution-relative propagation and would divide out any dam-induced compression of flow variance. Two direct checks show the streamflow null is not a standardization artifact. First, the buffering models and the upstream placebo are re-estimated on log 12-month mean flow, a volumetric semi-elasticity (percent of flow per SPEI unit) that no per-gauge normalization can compress: all slope gaps remain permutation-null, with point estimates slightly positive (toward steeper, not buffered, transmission) and the upstream placebo at least as large as downstream, reproducing the siting pattern. Second, the compression premise itself: deseasonalized log-flow variability at regulated downstream gauges is statistically indistinguishable from unregulated upstream gauges (Wilcoxon p = 0.43), so there is no differential variance compression for the standardization to divide out at the monthly-to-annual scale our design resolves.

Quantity	Value	Detail
log-flow semi-elasticity gap, ITT (treat x SPEI)	0.0743	SE 0.0720, perm p = 0.562; baseline d log Q / d SPEI 0.139 (% flow per SPEI unit)
log-flow semi-elasticity gap, downstream (treat x SPEI)	0.0401	SE 0.0855, perm p = 0.782; baseline d log Q / d SPEI 0.099 (% flow per SPEI unit)
log-flow semi-elasticity gap, upstream placebo (treat x SPEI)	0.0952	SE 0.0682, perm p = 0.422; baseline d log Q / d SPEI 0.168 (% flow per SPEI unit)
deseasonalized log-flow SD by gauge class (relative variability)	0.8270	downstream 0.827 (n=61) vs upstream 0.902 (n=35) vs control 0.579 (n=121); up-down Wilcoxon p = 0.43

Supplementary Table S20 | Metric and functional-form checks. Top: the baseline SPEI-to-SSI transmission slope among undammed controls varies smoothly and monotonically across aridity terciles, with no regime break for the linear log-aridity adjustment to misfit. Middle: the control elevation sensitivity of transmission shows no detectable nonlinearity across the ~2000 m snowline (piecewise interaction p = 0.25); the snow regime itself is tested directly on the placebo estimand in Supplementary Table S12. Bottom: the cropland null is metric-invariant: re-fit on log cropland share (relative rather than absolute expansion), the forcing-DiD slope gap remains permutation-null.

Quantity	Value	Detail
control baseline SPEI->SSI slope by aridity tercile (arid / mid / humid)	0.6470	mid 0.699, humid 0.743; smooth monotone variation, no regime break
control elevation sensitivity of transmission, below vs above the ~2000 m snowline	- 0.0780	above 0.212 per km; interaction p = 0.25 (n=84 control gauges)
relative cropland expansion: forcing-DiD slope gap on log cropland share	0.0473	SE 0.046, perm p = 0.369 (140 units with nonzero cropland); metric-invariant null

Supplementary Table S21 | Evapotranspiration outcomes (relegated from the main text). The ET pathway is reported here with its caveats and carries no evidential weight in the paper's conclusions. Whole-basin ET fails its event-study pre-trend test (differential pre-2010 trend p approximately 3e-5) and is confounded by residual aridity over barren land; masking barren pixels removes both the pre-trend failure and the positive effect (Supplementary Fig. S7), but the resulting orchard stratum was defined post hoc and relies on MOD16, whose documented underestimation of ET over irrigated land biases its slope gap toward zero, disqualifying a null as evidence in a null-finding design. A thermal ET product (Landsat SSEBop/METRIC or ECOSTRESS class) is the way to test the consumptive-use pathway properly and is flagged as future work.

Group	Outcome	Estimand	Estimate	95% CI	p
Cross-sectional matched ATT	Reservoir ET buffering	d log ET / d SPEI	-0.014	[-0.0646, 0.0369]	-
Forcing-interacted DiD (slope-gap)	Whole-basin ET	differential deficit->impact slope	0.048	[0.0255, 0.0703]	0.11
Forcing-interacted DiD (slope-gap)	Orchard ET	differential deficit->impact slope	-0.020	[-0.0448, 0.00565]	0.24